“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 26

Any Reminders? Any Questions?

- Teaching evaluations are open!

- Exam 3: Friday May 1 – Ch 13.1-13.3 and 16.1-16.8

- Office hours Mondays 1-2pm online
- Review for Exam 3 during class on Thursday!
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

ALVARO: Start the recording!

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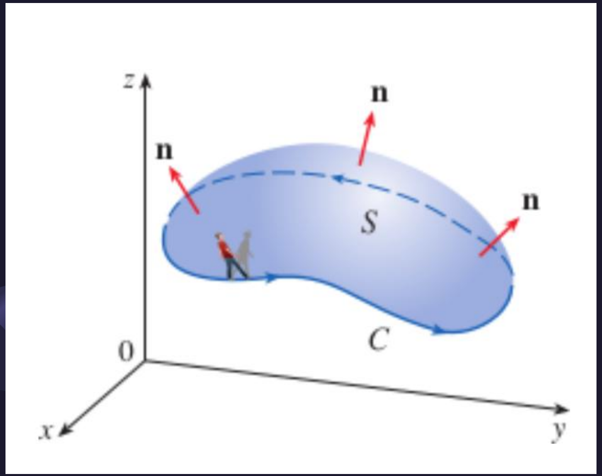
Stokes’ Theorem

Stokes' Theorem

Stokes' Theorem

Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S}$$



$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

$$= \operatorname{curl} \mathbf{F}$$

Sir
George Stokes
FRS

Stokes, 1860s

35th President of the Royal Society

In office
1885–1890

Preceded by	Thomas Huxley
Succeeded by	Lord Kelvin

Personal details

Born	George Gabriel Stokes 13 August 1819 Skreen, Ireland, United Kingdom of Great Britain and Ireland
Died	1 February 1903 (aged 83) Cambridge, England, United Kingdom of Great Britain and Ireland

Example: Evaluate $\int_C F \cdot dr$ where $F = (-y^2, x, z^2)$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. (C is oriented counterclockwise when viewed from above.)

Example: Evaluate $\iint_S curl(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy-plane.

Alternative solution: Change S to a disk!

[Extra Space]

Example: Evaluate $\iint_S curl(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy-plane.

Alternative solution: Change S to a disk!

Example: Evaluate $\iint_S \ curl(F) \cdot dS$ where $F = (2y\cos(z)\,, e^x\sin(z), xe^y)$ and S is the semisphere $x^2 + y^2 + z^2 = 9,\ \mathrm{z} \geq 0$ and oriented upwards.

[Extra Space]

Example: Evaluate $\iint_S \ curl(F) \cdot dS$ where $F = (2y\cos(z), e^x \sin(z), xe^y)$ and S is the semisphere $x^2 + y^2 + z^2 = 9, \ \mathrm{z} \geq 0$ and oriented upwards.

Example: Evaluate $\iint_S \ curl(F) \cdot dS$ where $F = (2y\cos(z)\,, e^x\sin(z), xe^y)$ and S is the cylinder $x^2 + y^2 = 9,\ 0 \leq z \leq 4$ along with the top at z = 4.

[Extra Space]

Example: Evaluate $\iint_S curl(F) \cdot dS$ where $F = (2y\cos(z), e^x \sin(z), xe^y)$ and S is the cylinder $x^2 + y^2 = 9,\ 0 \leq z \leq 4$ along with the top at z = 4.

Questions?

ALVARO: Start the recording!

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The Divergence Theorem

The Divergence Theorem

$$\text{div}\,\mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

The Divergence Theorem

Let E be a simple solid region and let S be the boundary surface of E, given with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E. Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \text{div}\,\mathbf{F}\,dV$$

> The Divergence Theorem is sometimes called Gauss's Theorem after the great German mathematician Karl Friedrich Gauss (1777–1855), who discovered this theorem during his investigation of electrostatics. In Eastern Europe the Divergence Theorem is known as Ostrogradsky's Theorem after the Russian mathematician Mikhail Ostrogradsky (1801–1862), who published this result in 1826.

Carl Friedrich Gauss

Portrait by Christian Albrecht Jensen, 1840 (copy from Gottlieb Biermann, 1887)[1]

Born	Johann Carl Friedrich Gauss 30 April 1777 Brunswick, Brunswick-Wolfenbüttel
Died	23 February 1855 (aged 77) Göttingen, Hanover

Example: Find the total flux of the vector field $F = (x^2, 0,0)$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

$x = a \sin\phi \cos\theta \qquad y = a \sin\phi \sin\theta \qquad z = a \cos\phi$

[Extra Space]

Example: Find the total flux of the vector field $F = (x^2, 0, 0)$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

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Example: Find the total flux of the vector field $F = (z, y, x)$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

$x = a \sin\phi \cos\theta \qquad y = a \sin\phi \sin\theta \qquad z = a \cos\phi$

[Extra Space]

Example: Find the total flux of the vector field $F = (z, y, x)$ over the unit sphere $x^2 + y^2 + z^2 = 1.$

$$x = a \sin\phi \cos\theta \qquad y = a \sin\phi \sin\theta \qquad z = a \cos\phi$$

Example: Evaluate $\iint_S F \cdot dS$ where F is the vector field $F = (xy, y^2 + e^{xz^2}, \sin(xy))$ and S is the surface of the region bounded by $z = 1 - x^2$ and planes z = 0, y = 0, and y + z = 2.

[Extra Space]

Example: Evaluate $\iint_S F \cdot dS$ where F is the vector field $F = (xy, y^2 + e^{xz^2}, \sin(xy))$ and S is the surface of the region bounded by $z = 1 - x^2$ and planes z = 0, y = 0, and y + z = 2.

Example: Find the flux $\iint_S F \cdot dS$ where F is the vector field $F = (0, y, -z)$ and S is the surface of the region bounded by $y = x^2 + z^2$, $0 \le y \le 1$ together with $x^2 + z^2 \le 1$, $y = 1$.

[Extra Space]

Example: Find the flux $\iint_S F \cdot dS$ where F is the vector field $F = (0, y, -z)$ and S is the surface of the region bounded by $y = x^2 + z^2$, $0 \leq y \leq 1$ together with $x^2 + z^2 \leq 1$, $y = 1$.

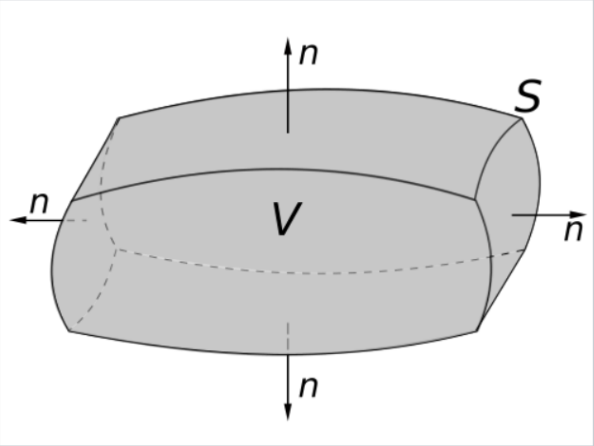
The Divergence Theorem (Gauss's Theorem)

Mathematical statement [edit]

Suppose V is a subset of $\mathbb{R}^n$ (in the case of $n = 3$, V represents a volume in three-dimensional space) which is compact and has a piecewise smooth boundary S (also indicated with $\partial V = S$). If $\mathbf{F}$ is a continuously differentiable vector field defined on a neighborhood of V, then:[4][5]

$$\iiint_V (\nabla \cdot \mathbf{F})\, \mathrm{d}V = \oiint_S (\mathbf{F} \cdot \hat{\mathbf{n}})\, \mathrm{d}S.$$

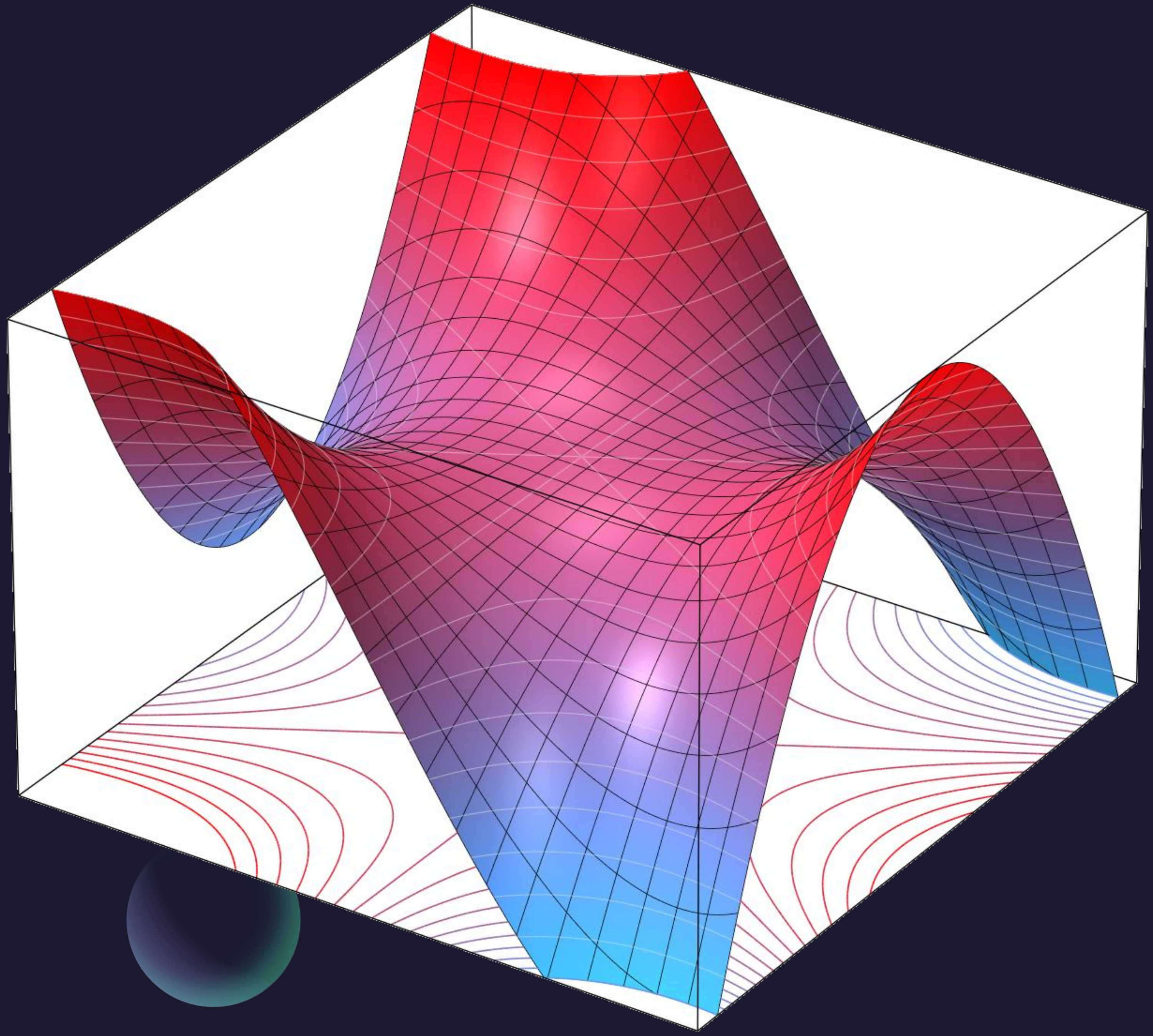
The left side is a volume integral over the volume V, and the right side is the surface integral over the boundary of the volume V. The closed, measurable set ∂V is oriented by outward-pointing normals, and $\hat{\mathbf{n}}$ is the outward pointing unit normal at almost each point on the boundary ∂V. ($\mathrm{d}\mathbf{S}$ may be used as a shorthand for $\mathbf{n}\mathrm{d}S$.) In terms of the intuitive description above, the left-hand side of the equation represents the total of the sources in the volume V, and the right-hand side represents the total flow across the boundary S.



A region V bounded by the surface $S = \partial V$ with the surface normal n

Thank you

Until next time.



Questions?

1. Vector Functions and Curves

P1. Find the unit tangent vector $\vec{T}(t)$ for $\vec{r}(t) = \langle t, \frac{2}{3}t^{3/2}, t^2 \rangle$ at $t = 1$.

P2. Find the length of the curve $\vec{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ for $0 \le t \le \ln(2)$.

P3. Consider the curve C formed by the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1 - y$.

(a) Find a parametrization $\vec{r}(t)$ for C.

(b) Find $\vec{r}\,'(t)$ and the line integral $\oint_C \langle -y, x, z \rangle \cdot d\vec{r}$.

2. Conservative Fields and Green's Theorem

P4. Use Green's Theorem to evaluate $\oint_C (xy + e^{x^2})\,dx + (x^2 + \sin(y^2))\,dy$ where C is the boundary of the rectangle with vertices $(0,0)$, $(3,0)$, $(3,2)$, and $(0,2)$.

P5. Show that $\vec{F} = \langle y\cos(xy), x\cos(xy) + 2y \rangle$ is conservative and find a potential function f. Use it to evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0,0)$ to $(1,\pi)$.

3. Parametric Surfaces and Surface Area

P6. Let S be the part of the plane $3x + 2y + z = 6$ in the first octant.

(a) Find a parametrization $\vec{r}(x,y)$ for S.

(b) Express the domain D as a **Type I (Type-x)** domain.

(c) Compute $\vec{r}_x \times \vec{r}_y$ and find the surface area of S using integration.

P7. Find the area of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 4$.

4. Flux and Vector Integrals

P8. Let S be the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 2$.

(a) Write a parametrization $\vec{r}(r, \theta)$ for S using **polar coordinates**.

(b) Find the normal vector $\vec{r}_r \times \vec{r}_\theta$.

(c) Evaluate the flux of $\vec{F} = \langle x, y, z^4 \rangle$ through S with upward orientation.

5. Stokes' and Divergence Theorems

P9. Let $\vec{F} = \langle z^2, x^2, y^2 \rangle$. Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where C is the triangle with vertices $(1, 0, 0), (0, 1, 0), (0, 0, 1)$, oriented counterclockwise when viewed from above.